Missing fossils as evidence: what the rocks can say about the age of land plants

Barry W. Brook*[1], Claude Opus 5.5 (Anthropic) and OpenAI Codex[2]

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1. University of Tasmania, Hobart, Tasmania, Australia
2. AI research collaborators; roles are stated in the Declaration of AI use

Corresponding author: Barry W. Brook

Abstract

Molecular clocks date living land plants' ancestor from the Cambrian to Late Ordovician, partly through differing fossil maxima. We examined three: 515.5 million years ago (Ma), resting on qualitative argument, and 459.1 Ma for living land plants; 431.2 Ma for living vascular plants. No three-rayed (trilete) spores are reported in Ordovician Libyan borehole samples, most from stages with such spores elsewhere: absence can be local. For vascular plants, the rule of three applies: if fossils turn up at a steady rate after the oldest one, the silence before it fits any earlier history expected to yield about three finds or fewer at that rate. The fewer finds that are independent discoveries, the longer three take; at a steady rate the 95% limit on the record's start is 428.8 Ma if all 48 are, near 431.2 Ma if half are. Ancestors at 444–452 Ma fit if early fossils were found 2.4–12 times less often, depending on that count, if finding rose fast enough, or if recognisable bodies appeared later. Finding proportional to a post hoc product of North American rock area and trilete-reporting share would fit the early silence with all 48 finds independent, but predicts 0.1 late-Silurian finds against 13.

Keywords: molecular clock; fossil calibration; land plants; vascular plants; stratigraphic confidence interval; Ordovician–Silurian

Introduction

Molecular clocks estimate when lineages diverged by combining DNA differences with fossil limits on the ages of selected ancestors. A crown group comprises the most recent ancestor shared by its living members and all its descendants, including extinct lineages. Clocks date these crown ancestors; extinct lineages outside the crown may have reached land earlier. For the ancestor of all living land plants (crown embryophytes), Morris et al. [1] obtained 515–474 Ma across seven tree topologies and Harris et al. [2] 515–493 Ma. Hooper et al. [3] reanalysed Harris et al.'s DNA data with revised fossil calibrations and obtained 457–450 Ma, a mean 55 Myr younger. For the ancestor of living vascular plants (crown tracheophytes: clubmosses and their relatives, ferns and seed plants), the three studies give 451–419, 452–447 and 432–428 Ma (figure 1); Su et al. [4] obtained older land-plant ages still, under a Proterozoic maximum. The dates bear on the order in which plant characters were assembled, and on how early plants changed productivity, weathering and climate [5–8].

A fossil sets a minimum age: its lineage existed by the age of its oldest secure member. A maximum based on absence adds an argument that older rocks would probably have preserved and yielded recognisable remains had the lineage existed [9]. How calibrations are specified can strongly shape estimated dates, and with large sequence datasets they can dominate the uncertainty in absolute ages [10,11]. A soft maximum permits older ages through a specified tail of the calibration prior [12]. Whether absence can set a maximum is contested [13,14], and best practice asks each calibration to justify its maximum from the relevant geological, geographical and taphonomic record [15].

Linking a fossil to an ancestor crosses two intervals: from divergence to the first recognisable character, and from that character to its first recorded fossil [16]. Both are uncertain for early land plants. Dispersed spores appear tens of millions of years before branched plant bodies [17–19], terrestrial deposits are scarce before the latest Silurian, at least in western Europe [5], and the oldest parent plants known in detail are minute [20].

Three maxima divide the current clocks (table 1). Morris et al. set 515.5 Ma for living land plants, arguing that older marine rocks rich in algal cysts would also have yielded land-plant spores [1,14]. Hooper et al. set 459.1 Ma from the oldest possible trilete (three-rayed) spores, in the Borenshult-1 core of southern Sweden [21], and 431.2 Ma for living vascular plants from Irish *Cooksonia* [22]. The three therefore rest on a preservation analogy, on dispersed spores and on plant bodies, each with its own route into the record, and each needs its own test.

We ask three questions of the published record. What evidence supports each maximum, and what links connect it to the dated ancestor? What can the absence of spores from one Libyan basin show? How much less often would early vascular-plant fossils have to be found, or how long after the ancestor would recognisable bodies have to appear, for older dates to fit? Established theory sets an expectation for each. An argument from absence needs documented chances for remains to be delivered, preserved and found [15,16]. Where older and younger rocks were sampled comparably, each further sample reporting no target spores tightens the limit on a high, steady spore supply. With the oldest find and the time window fixed, more independent finds bring the limit on a record's start closer to that find, and a lower finding rate before it pushes the limit back [23,24]. The audit identifies what each maximum assumes, the Libyan record shows how far a spore absence reaches, and the body-fossil models describe the finding rates or delays that would fit older vascular-plant dates.

Table 1. The three maxima examined: the evidence each rests on, the test the published record allows, and the links a clock must still assume. GTS2012, Geologic Time Scale 2012.

Maximum	Set by	Evidence	Test available	Links assumed
515.5 Ma, living land plants	Morris et al.	Base of the *Olenellus* Biozone, below the oldest possible Cambrian spore-like packets; older marine rocks with algal cysts	Affinity of the Cambrian packets; any fully reported older sequence	Spore delivery to marine basins, catchment, search effort, spore production by the earliest members

Maximum	Set by	Evidence	Test available	Links assumed
459.1 Ma, living land plants	Hooper et al.	GTS2012 Sandbian base plus uncertainty, keyed to the oldest possible trilete spore (Borenshult-1, Sweden)	A Libyan worked case of local absence, compared with records from overlapping stage ranges elsewhere	Earliest crown members produced triletes, the trait persisted, producers lived in the catchment
431.2 Ma, living vascular plants	Hooper et al.	GTS2012 Homerian base plus uncertainty, keyed to Irish Homerian *Cooksonia*	A limit on when the body-fossil record began, and what older ancestors require	Finding rate before the oldest find, delay before recognisable bodies, the oldest finds lie within the vascular crown

Material and methods

(a) Timescale and interpretations. Analysed ages use the stage bases of the International Chronostratigraphic Chart, v2024/12 [25]; literature ages were mapped by stage. Published maxima keep their source timescale, GTS2012, so they can be compared as their authors set them. Each analysis declares which interpretation of disputed identifications it uses, main or inclusive, so contested fossils enter as alternatives.

(b) Source audit. For each maximum we traced the calibration fossil, its age rule and how it was found back to the original descriptions, and recorded each published interpretation of the fossil's identity with its evidence [15]. A reported absence applies to the material examined, and a missing report counts as an absence where the source reports the target fully.

(c) Libyan spore records. Abuhmida [26] reported spores and other microfossils from Ordovician to lower Silurian wells in the Murzuq Basin, southwest Libya. We transcribed all 164 appendix samples with their chart and prose reports. The analysis uses the chart-coded samples of drill cuttings (rock chips returned during drilling) from five wells: three supply 28 samples older than the Llandovery (earliest Silurian), and all five supply 59 Llandovery samples, which serve as the reference. The source states the absence of trilete spores explicitly for part of the sequence; for the Hawaz Formation, samples reporting none rest on complete reporting charts and the stated microscopy protocol. The basin lay near the Ordovician South Pole, within the area of Hirnantian glaciation [27,28] (electronic supplementary material, figure S2). We compared the samples' ages with trilete records in our spore compilation, built from Palynodata [29], the curated assemblages of Wellman et al. [30] and later literature, keeping records with usable ages that are not flagged as reworked or questionable. A model of spore supply, in which samples from one well share a well effect, sets upper limits on how much steady supply the samples allow; its 95% limits were tested on simulated data with known answers (electronic supplementary material, S2).

(d) Body-fossil finds. We assembled land-plant body fossils dated to 453–393 Ma from Paleobiology Database (PBDB) occurrences, whose names an AI model classified with cited sources before our audit, and from a source-reviewed set of the oldest records with their competing interpretations. Collections, taxa or papers sharing a locality and age count as one find, a proxy for an independent fossil horizon. We declare three nested categories. Land-plant bodies are any accepted macro- or mesofossil. Polysporangiophytes have documented branched axes bearing sporangia (spore capsules), or placement within the vascular crown. Crown vascular plants belong to the lycophyte line (clubmosses and relatives) or the euphyllophytes (ferns and seed plants) [31–34]. Under the main interpretation the three categories hold 78, 71 and 50 finds, of which 76, 69 and 48 are certainly older than the end of the Emsian (393.47 Ma), the younger edge of the time window (electronic supplementary material, S5.1).

(e) Limit on when the record began. Let $w(s)$ be a fixed relative rate at which recognisable fossils of

a category are found at age s, and $U(a) = \int_c^a w(s)\,ds$ the cumulative finding rate between a younger cutoff c, here the end of the Emsian, and age a. For n independent finds with oldest age A, the 95% confidence limit T on the start of the recognisable record solves $U(T) = U(A)\,0.05^{-1/n}$ (figure 2). With constant w this is the classical confidence interval on a stratigraphic range, and the absolute rate cancels [23,24]; varying w follows the extensions to unevenly spaced fossil horizons [35,36]. For finds dated by age ranges, A is the oldest possible age of the oldest find, and n counts only finds certainly inside the window; both choices move the limit older. Finds dated by stage may record one discovery more than once, so we report the limit as a function of the number of independent finds, with the main count, a count of region–stage cells and the curated oldest records as reference points. The limit concerns the start of the record; the Discussion treats its link to ancestors.

(f) Finding-rate histories and shape tests. We compared six assumed histories of the finding rate: steady; proportional to PBDB land and marginal-marine collections per Myr [37], with or without the collections that hold plant fossils; declining exponentially into the past with 25 or 10 Myr timescales; and plant-excluded collection density multiplied by the 10 Myr decline. The collection records reach back to 469.4 Ma, the oldest age the analysis can consider. For each history, an exact Kolmogorov–Smirnov test compared the spread of fossil ages it predicts with the observed spread, given the count and the oldest find. Because many ages are known only as ranges, we searched every possible assignment of ages within those ranges, including whether the finds straddling the cutoff fall inside the window, for the best fit [compare 38]. A history that fails under every assignment is rejected (electronic supplementary material, S5.2).

(g) Lower earlier finding rates and delays. The main scenario holds the finding rate steady after 426.7 Ma, the oldest possible age of a crown vascular-plant find, and lowers it before that age. For a candidate ancestor age t and an assumed delay d before the first recognisable bodies, the ratio of earlier to later finding rate that puts $t - d$ where the limit of (e) would fall is $r = (A-c)(0.05^{-1/n} - 1)/(t - d - A)$, defined where $t - d > A$; fossils would then have to be found $1/r$ times less often before the oldest find than after it. The longer the delay, the smaller the reduction a given ancestor age needs [16]. If fossils turn up at a steady rate after A and recognisable bodies appear with the ancestor (zero delay), a candidate age t fits the oldest find (an oldest find no older than A has probability at least 0.05) when the relative finding rate summed over [A, t] is at most $K = (A-c)(0.05^{-1/n} - 1)$. Writing R for that sum, each of the n finds falls before A with probability $p = R/(A - c + R)$; at the boundary $R = K$, no early find has probability $(1-p)^n = 0.05$, so $p \approx 3/n$: the rule of three for zero counts [39]. At the later rate estimated as $n/(A-c)$ finds per Myr, K corresponds to $np/(1-p) = n(0.05^{-1/n} - 1)$ expected finds before the oldest find, which approaches $\ln 20 \approx 3$ as n grows. The step is one shape among many; we also report a linear rise from zero at the ancestor and an exponential rise, each reaching the later level at A. We chose where to put the step and the rises after seeing the data, so these curves show what would fit, without the 95% confidence that (e) gives each fixed history.

(h) Outside measures. To judge whether a required reduction is large, we compared it with three rough measures, defined before they were computed: the area of North American Macrostrat columns holding land-laid (non-marine) rock [40]; PBDB land and marginal-marine collections per Myr [37]; and the share of Palynodata studies [29] that report trilete spores. Each measure's mean over the 33 Myr after the oldest find, divided by its mean from the oldest find back to a candidate ancestor age, gives how many times scarcer it was earlier, at zero delay. The measures are guides; they enter no 95% inference (electronic supplementary material, S5.8). Each covers the same two windows as the finding-rate requirement, so the two ratios can be compared in size. After seeing the single measures, we also multiplied the rock and spore measures stage by stage, as a further guide without confidence level, and asked whether the later finds lie where each measure is large: a measure that finding followed stage by stage would put most later finds where it is high. We kept searching out of the product, because collections are made in rock and the product would count rock twice, and multiplied it with the spore measure instead as a global alternative.

Results

(a) The 515.5 Ma maximum rests on a qualitative argument. Morris et al. placed this maximum below the oldest possible total-group records, spore-like packets and clusters from the Cambrian Stage 4 Rome Formation, at the base of the *Olenellus* Biozone (about 515.5 Ma). Their preservation argument, from older marine rocks with algal cysts and without land-plant spores, shows that walled microfossils survived

and were recovered [1,14]. How many spores reached those basins, the land areas they drained and the search effort remain unmeasured in the sources we assessed. All three sources we examined assign the Cambrian packets to algae or lineages before land plants: Charophyta [41], a pre-embryophytic grade [42] or ancestral aeroterrestrial algae [43] (electronic supplementary material, S4).

(b) The younger maxima follow a stage-base rule. Hooper et al. set both younger maxima as a GTS2012 stage base plus its uncertainty. They keyed 459.1 Ma (458.4 + 0.7 = 459.1 Ma) to a possible trilete spore at 48.2 m in the Borenshult-1 core [21]. The oldest definite trilete level, at 43.1 m, is also Sandbian and lies below the 454.41 Ma Kinnekulle K-bentonite, so the rule gives the same value from either level (electronic supplementary material, S2.4). They keyed 431.2 Ma (430.5 + 0.7 = 431.2 Ma) to Irish Homerian *Cooksonia* [22]. The older Czech *Cooksonia barrandei*, a branched sporophyte with terminal sporangia from a Sheinwoodian (early Silurian) horizon dated from its slab fauna and by correlation [32], meets the same criterion (electronic supplementary material, table S13). On the current timescale the rule gives outer values of 431.9 Ma for the Homerian and 434.1 Ma for the Sheinwoodian.

(c) The Libyan samples show that a spore absence can be local. Trilete spores are reported in 51 of 59 Llandovery samples and in none of the 28 older samples (electronic supplementary material, figure S2). Of the older samples, 18, from three wells, are Katian–Hirnantian (Late Ordovician). They are younger than the definite Swedish triletes below the 454.41 Ma bentonite, and fall in the same stages as trilete-bearing assemblages in Saudi Arabia and Turkey [[19]; [44]][] (electronic supplementary material, S2.6). 5 of the Libyan samples, and a Saudi assemblage, are Hirnantian, a stage of about two million years. Trilete spores thus occur elsewhere in the stages these Libyan rocks represent, and none is reported from them. Only 10 broadly dated Darriwilian (Middle Ordovician) samples, from two wells, have age ranges reaching beyond 459.1 Ma, and their weight depends on how ages are spread within the stage. With equal delivery, the samples limit steady trilete supply since 459.1 Ma to 0.091 of the Llandovery level under the main model, and to about 0.4 when samples are weighted by well and rock unit (electronic supplementary material, S2.2 and S2.5). Lower delivery, lower local supply or producers outside the catchment all fit.

(d) The silence before the oldest find fits about three expected finds, so its limit depends on how many finds are independent. Every calculation in which fossils turn up at a steady rate after the oldest find shares one rule. Take that rate, estimated from the count: the absence of older finds fits any earlier history expected to yield about three finds or fewer at that rate, from 3.09 with 48 independent finds to 3.32 with 15 (Material and methods (g)). The count sets how long three finds take at the later rate: 2.1 Myr with 48 finds, 5.7 Myr with 19 and 7.3 Myr with 15. At a steady finding rate, the 95% confidence limit on the start of the crown vascular-plant record lies that far before the oldest find (figure 3a). Counting the 48 finds certainly inside the window as independent gives 428.8 Ma; treating each region and stage as one find (19) gives 432.4 Ma, and counting only the curated oldest records (15) gives 434.05 Ma. The three share the oldest find and the cutoff, so they lie on one curve, which crosses the 431.2 Ma maximum at about 24 independent finds, half the main count. The main count already merges repeated collections from one locality and age; the region-and-stage count treats every locality in a region and stage as one discovery, and the curated count keeps only the source-reviewed oldest records. These are counting scenarios, not estimates of the number of independent finds; the curve shows which count each limit assumes. Coarse stage ages are one reason the count is uncertain: finds from one region and stage may come from one bed or from beds millions of years apart. Over-counting is a known way for the method to fail. In simulations in which every true find was counted twice, the 95% limit held in only 0.78 of trials (electronic supplementary material, S5.6). If the 48 records instead represent 24 independent paired finds, their conditional steady-rate limit is near the 431.2 Ma maximum.

With all 48 finds independent, finding rates that track collection density give 428.9 Ma, or 428.5 Ma with plant-bearing collections removed, and a 25 Myr decline into the past gives 431.6 Ma. If finding fossils became steeply harder with age (the 10 Myr decline), the method sets no limit as far back as 469.4 Ma, the edge of our data. That history, however, predicts a spread of fossil ages unlike the one observed, however the uncertain ages are assigned. The 25 Myr decline passes the shape test under one assignment of ages when all 50 possible finds are counted ($p = 0.067$), and fails under every assignment of the 48 finds certainly inside the window (best $p = 0.035$). Shifting stage boundaries by their quoted uncertainties, in the scenarios tested, moves the main limit between 427.31 and 430.42 Ma (electronic supplementary material, S5.3).

(e) Older vascular-plant dates fit if early fossils were found less often, or recognisable bodies appeared later. In the main scenario, halving the finding rate before the oldest find moves the limit to 431.0 Ma, still younger than the 431.2 Ma maximum. Ancestors at 444–452 Ma, spanning the Morris et al. mean dates and the Harris et al. interval, fit if fossils were found 8–12 times less often before the oldest find than after it, with recognisable bodies appearing with the ancestor and all 48 finds independent (figure 3*b*). These factors spread three finds' worth of finding over the 17–25 Myr between the ancestor and the oldest find. With about 24 independent finds, 3.9–5.7 times suffices, and with 19 or 15, 2.4–4.5 times; the reduction needed grows roughly in proportion to the number of independent finds, and with fewer than about 24, 431.2 Ma already fits a steady rate. At a steady finding rate and with all 48 finds independent, the same ancestors fit if recognisable bodies appeared about 15–23 Myr or more after them, leaving at most the 2.1 Myr that three finds take; fewer independent finds shorten the delay needed. Smaller reductions combine with shorter delays (figure 3*b*).

(f) The record limits early finding to about three finds in total, whatever its shape. With fossils turning up at a steady rate after A and zero delay, a straight-line rise from zero at the ancestor stays within the three-find total of (*d*) only for ancestors up to 431.0, 438.1 and 441.4 Ma with 48, 19 and 15 finds. An exponential rise fits ancestors at 444–452 Ma if finds doubled about every 1.5 Myr or faster with 48 finds, every 4.0–4.2 Myr with 19 and every 5.3–5.8 Myr with 15. As candidate ages grow older, the permitted doubling time approaches a count-specific threshold, $K \ln 2$, about 1.48, 3.93 and 5.09 Myr; a rise this fast or faster fits an ancestor of any age, because its whole early tail sums to about three finds or fewer. These are requirements for fitting. For coarse context, the share of palynological studies reporting triletes changed between the Homerian and the Gorstian, across the oldest find, at an equivalent doubling time of about 6.6 Myr, and rose fastest, about every 1.0 Myr, from the Hirnantian into the Rhuddanian; both give scale only.

(g) Multiplied, the rock and spore measures would allow older ancestors if finding shared their contrast between the windows. For ancestors at 444–452 Ma, the mean area of North American columns holding non-marine units is about 3.2–3.5 times lower before the oldest find than after it, and PBDB land and marginal-marine collections are only 1.35 and 1.41 times sparser at 444 and 452 Ma. The share of indexed palynological studies reporting trilete spores is 5.1–7.1 times lower (figure 3*b*; electronic supplementary material, S5.8). The rock ratios are numerically as large as the required finding-rate ratios for about 14–20 or fewer independent finds, and the spore-reporting ratios for about 29–31 or fewer. With all 48 finds independent, if body-fossil finding were proportional to each measure in both windows, about 7.6–11.5 finds would have been expected before the oldest find under the rock measure and 4.9–5.1 under the spore-reporting share, against about three. South American rock, recorded more coarsely, shows no decline.

Multiplying the two measures, a combination we chose after seeing both, brings early finding within the rule. Taken stage by stage, their product is 21.7–29.8 times lower before the oldest find than after it for ancestors at 444–452 Ma, leaving 1.15–1.23 expected early finds with all 48 finds independent, or 1.48–1.61 if the two ratios are multiplied directly; every count fits. The global searching measure multiplied by the spore share leaves 3.6–4.1 expected early finds with 48 finds and 1.8–2.0 with 24. The case for multiplying is that rock stands for where bodies could be preserved and collected, the spore share for how common trilete producers were, and a find needs both. The case against is that the two may share a cause and count one effect twice; the spore share counts studies, and trilete producers include plants outside the crown group; and the rock is North American, while the oldest finds come from Australia, Arctic Canada and other regions. Across the 13 stages from the Katian to the Emsian, the rock and spore series have a weak rank correlation (Spearman $\rho = 0.30$, permutation $p = 0.31$), within the range chance gives; both are high in the Early Devonian, which makes the stage-by-stage product larger than the product of the two ratios. So few stages cannot separate a shared cause from coincident trends.

The later window tests the proportional reading directly. Of the 48 finds, 13 date wholly to the Pridoli or the Ludfordian–Pridoli. Finding proportional to the rock measure would put 0.45 finds in the Gorstian–Pridoli, the spore share 3.8 and their product 0.14, each with one-sided binomial p below 0.003; counting by region and stage, or starting the interval at the Ludfordian, gives the same result. A level rate puts 10.2 finds there and searching 10.1, both compatible with the 13 that certainly lie there, though up to 16 may; from the Ludfordian base each puts about 7.8 ($p = 0.04$, an interval set after the finds' ages were seen), so the later finds sit somewhat more in the late Silurian than a level rate predicts. The later record therefore

contradicts finding that follows the rock or spore measure stage by stage. Using their combined contrast for older ancestors needs a separate assumption: that mean finding rates share that contrast between the two windows. A delay before recognisable bodies remains another route.

(h) One Ordovician find moves the broader land-plant limit. Sporangial fragments with in situ cryptospores from the Ghaba-1 borehole, Oman, dated to the Katian on later correlation [45], form one land-plant body find. Adding them moves the steady-rate limit for all land-plant body fossils from 434.5 to 455.2 Ma, and the enlarged set fails the shape test under all six histories. The fragments came from palynological preparation of borehole material, whereas most later finds are compressions collected from bedding planes [46].

Discussion

The early record allows older ancestors, and the analyses state what each would require. The evidence differs in depth between the two clades. For living land plants it comes from the audit of the 515.5 Ma maximum and one polar basin, whose samples report no triletes from stages in which triletes occur elsewhere. Those samples exclude a steady, well-delivered supply near the Llandovery level, while lower delivery, lower local supply or spore producers outside the catchment all fit. For living vascular plants it comes from a body record drawn from several regions: at a steady finding rate, the 95% confidence limit on the start of the recognisable crown vascular-plant record lies between 429 and 434.05 Ma, depending on how many finds are independent, and passes the 431.2 Ma maximum at about half the main count. For each way of counting, a lower earlier finding rate, later recognisable bodies or both fit ancestors older than these limits; the models give the size of each requirement in aggregate. If fossils turned up at a steady rate after the oldest find and recognisable bodies appeared with the ancestor, the record allows about three finds' worth of earlier finding, so it limits the total of early finding, and every history with that total fits equally. A fast enough exponential rise keeps that total below three for an ancestor of any age, so an older limit from the body record needs an assumption that restricts the early finding history.

Several processes could have made early plants scarcer in the record. Producers may have been rarer, more geographically restricted, concentrated where little terrestrial sediment accumulated, or preserved in intervals that have been searched less. Body size, cuticle and lignified tissue could also raise the chance that a body is preserved, found and recognised. Branched sporophytes and conducting tissue were acquired stepwise along the lineages being dated [47]. The early branched sporophyte *C. barrandei* is preserved in marine sediments [32], and the Oman fragments show that small sporangia can survive [45]. The models give the combined reduction a candidate age requires. The outside measures show less column area holding land-laid rock in North America and a lower share of indexed palynological studies reporting trilete spores, which changing coverage, study aims, preservation and identification can also lower. These differences give the required ratios a scale. Multiplied, they would let even 48 independent finds fit ancestors at 444–452 Ma with zero delay, if mean finding rates shared their contrast between the windows. Finding that followed them stage by stage would have left almost none of the late-Silurian finds, so the record contradicts close tracking and leaves the shared contrast as a separate assumption, with no confidence level.

Silvestro et al. [48] modelled genus-level PBDB plant occurrences with a joint Bayesian model of preservation and diversification, and estimated the origin of vascular plants at a mean of 433.5 Ma (95% highest posterior density 449.0–424.0 Ma). They estimate when the earliest vascular-plant genus lineage arose, with a fitted, varying preservation rate; we limit the start of the recognisable record of a declared crown category under assumed finding-rate histories, then ask what older ancestors require. The two analyses target different quantities from partly shared data and test different assumptions. CladeDate builds calibration densities directly from fossil times [49], and Hooper et al. link spore-character reconstruction to revised calibrations. We make the maxima's observational requirements explicit, limiting local spore supply and sizing the reductions in finding rate, or the delays, that fit older vascular-plant dates.

Conditions. The Libyan limits assume that the Llandovery spores are indigenous, that older and younger samples were reported comparably, that ages are uniform within stages, and that samples are independent apart from a shared well effect; the four-tenths figure comes from an exploratory weighting that was not tested on simulated data (electronic supplementary material, S2.5). The comparison with other regions holds at stage resolution and depends on the coverage of the spore compilation. Linking the local spore record to

ancestors requires that the earliest crown embryophytes produced recognisable trilete spores, that the trait persisted and that producers lived in the catchment (electronic supplementary material, S6).

The body-fossil limits and shape rejections assume independent ages on a continuous scale; fixed finding-rate histories, cutoff, timescale, inventory and counting rules; and an observation process whose reported age ranges always contain the true ages, including whether each find falls inside the window. Geological age ranges carry no demonstrated joint coverage; if the probability that any range misses its true age is δ, the test's error rate is at most $\alpha+\delta$, where α is its level for exact ages. Simulations with exact ages gave coverage of 0.941–0.958 (electronic supplementary material, S5.6). The step scenario was chosen after seeing the data. Linking the vascular-plant limit to ancestors requires that the oldest finds belong to the crown group, and a specified delay before recognisable bodies. The rule of three counts expected early finds at the later rate estimated from the same finds, so it is conditional on the count and the cutoff, and a delay moves the start of that count to $t - d$. The rises and the window-mean reading of the outside comparison assume a level finding rate after the oldest find, which the spread of later fossil ages can test, while finding proportional to a measure lets the later rate vary with it; the doubling times are requirements, and the spread of early plants is unmeasured. The outside measures describe rock and published studies; each ratio would equal the ratio of mean finding rates if finding rate were proportional to that measure, with the same constant in both windows. The placement check contradicts finding proportional to the rock measure, the spore share or their product within the later window; a level rate and searching remain compatible with the Gorstian–Pridoli allocation at the minimum count, a result that depends on uncertain ages and the interval chosen. The comparison therefore assumes that mean finding rates share each measure's contrast between the windows. Each placement check tests one late-Silurian allocation, using the 13 finds (six region-and-stage units) dated wholly within it, while up to 16 may lie there, independent units and age ranges that contain the true ages, and inherits any imbalance in how the inventory sampled Silurian and Devonian records. The combined measure was chosen after the single ones were seen, has no confidence level, and may count a shared cause twice.

Published evidence could narrow these requirements. Records of searched Silurian terrestrial beds, with effort and empty beds included, would help constrain early finding rates, and rock-unit compilations beyond North America would test the rock measure [40]. A second complete spore sequence outside polar Gondwana, with older samples reporting no triletes, younger samples reporting them and counted target and control grains, would show whether supply limits of this size hold outside one polar basin; a fully reported marine sequence older than 515.5 Ma, with comparable recovery documented, would help quantify the oldest maximum's absence argument. Finer, independent ages for the Yea (Victoria, Australia) and Bathurst Island (Arctic Canada) horizons, which hold the oldest possible crown vascular-plant finds, would sharpen the body-fossil limit [31,33], and finer ages generally would show which finds share a bed, and so how many are independent. Characters placing the oldest branched sporophytes on the lycophyte or euphyllophyte side of the vascular crown could change which fossils enter the category, as well as their link to the ancestor.

Conclusion

A fossil maximum based on absence combines expectations about ancestral characters with the chance of finding their remains in older rocks. Clock builders can state each maximum's character, geographical and finding-rate assumptions, and justify the probability assigned to older ages against them [12,15]. Testing alternative defensible maxima then shows how much each assumption moves the timescale, and sampling the joint age prior without molecular data, then comparing its effective marginal ages with the posteriors, shows what the molecular data add [50].

For the oldest vascular-plant body fossils, the rule of three makes the finding-rate assumption concrete: if fossils turned up at a steady rate after the oldest find, the silence before it fits about three expected finds at that rate. An older ancestor needs early finding low enough, or recognisable bodies late enough, to keep that expected count at about three or fewer, and the number of independent finds sets how many million years three finds span. Multiplied, North American non-marine rock area and the share of studies reporting triletes fell enough before the oldest find for ancestors at 444–452 Ma at every count, with zero delay, if mean finding rates shared that fall; finding that followed them stage by stage is contradicted by the late-Silurian finds, so this route rests on that shared fall alone. For spores, the Libyan basin shows that an absence can be

local: older land-plant ancestors fit its record if, with equal delivery, their steady trilete supply since 459.1 Ma stayed below about a tenth of the later supply, with older onsets needing less, or if the producers lived elsewhere or acquired triletes later. Missing fossils thus allow older vascular-plant ancestors if early plants were found less often or became recognisable later, and the body record states how much less often.

Ethics

We analysed published descriptions and data. The study involved no fieldwork, specimen collection or research involving human participants.

Data accessibility

The repository https://github.com/BWBrook/silurian_plants holds the source-linked derived tables, model code, qualification evidence, body-record outputs with hash receipts, figure and table scripts, and the claims ledger that binds every reported number to its evidence. The frozen PBDB collections extract used by the effort histories is archived under CC0 [37]. Copyrighted full texts are excluded from the repository; source routes and held-file hashes are recorded separately. Openly licensed third-party deposits are fetched by script and hash-checked. Supplementary material is available online.

Declaration of AI use

Claude Opus 5.5 (Anthropic) developed and ran analyses and drafted and revised the manuscript. OpenAI Codex (`gpt-6.1-sol` and `gpt-6.1-terra`) curated and checked source data, implemented analyses, audited statistical methods and reproduction, and revised the manuscript. Earlier Codex work used `gpt-5.6-sol` and `gpt-5.6-luna` for source recovery and methodological checks. Claude and Codex reviewed each other's work. GPT-6 Pro (OpenAI) supplied independent critiques, which were checked against sources and saved outputs before integration. Claude Fable 5.2 (Anthropic) advised from 12 November 2026 through critiques and prompts relayed by Barry, including the proposal to organise the body-fossil results around the rule of three; it made no edits. Barry W. Brook originated and guided the programme and contributed scientific decisions; he will review the final manuscript before submission. Task-specific roles, model versions and the limits of sessions whose exact variant was unavailable are recorded in `docs/charter.md`, `studies/f3/docs/audits/` and `studies/f3/manuscript/reconstruction/`, with adopted decisions in `docs/decisions.md` and `studies/f3/docs/decisions.md`.

Authors' contributions

[Byline and final CRediT wording remain deferred to the submission phase. A contribution inventory is prepared in `studies/f3/manuscript/reconstruction/4-submission-materials.md`.]

Conflict of interest declaration

[Barry to confirm.]

Funding

This research received no specific funding.

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439 **Fig. 1.**

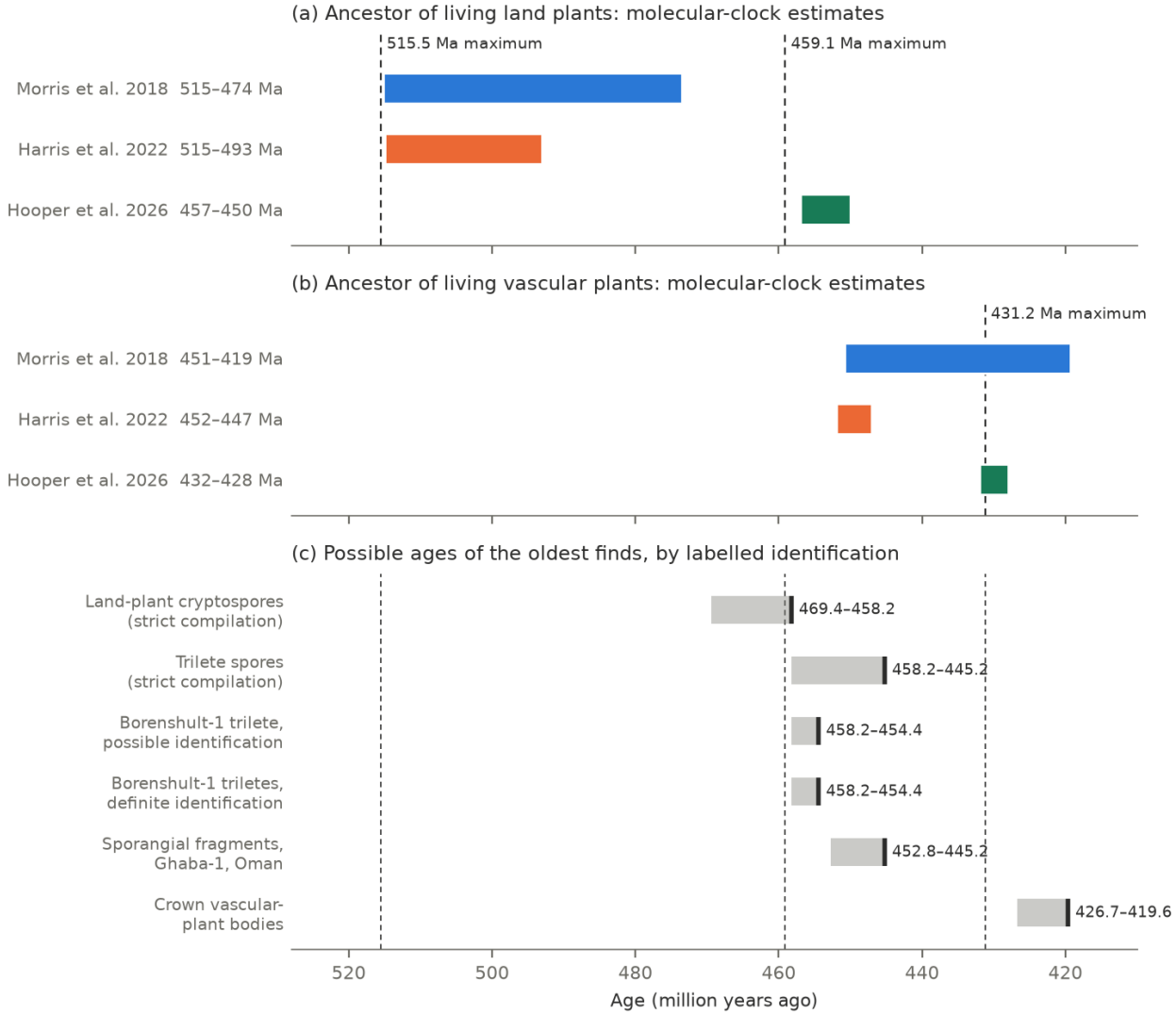


Figure 1: The dating dispute and its fossil evidence. Upper bars give dates for the ancestor of living land plants; the next group gives living vascular plants. Morris et al.'s bars envelope 95% highest posterior density (HPD) intervals across seven tree topologies; Harris et al. and Hooper et al. give their study-specific 95% HPD intervals [1–3]. Vertical lines mark the maxima of 515.5, 459.1 and 431.2 Ma. Clock estimates and maxima keep their published numerical ages. Fossil stage ages use the International Chronostratigraphic Chart v2024/12; independent radiometric ages keep their published values. Each fossil bar spans the possible ages of a category's oldest find, from earliest possible to earliest certain occurrence; the two ends may come from different finds. Separate bars show cryptospores and triletes under the strict compilation interpretation, possible and definite Borenshult triletes [21], Oman's sporangial fragments [45], and crown vascular-plant body fossils [31,33]. The two Borenshult interpretations share this coarse age bracket. Each age holds for the identification labelled; the queried specimen remains a possible trilete.

Fig. 2.

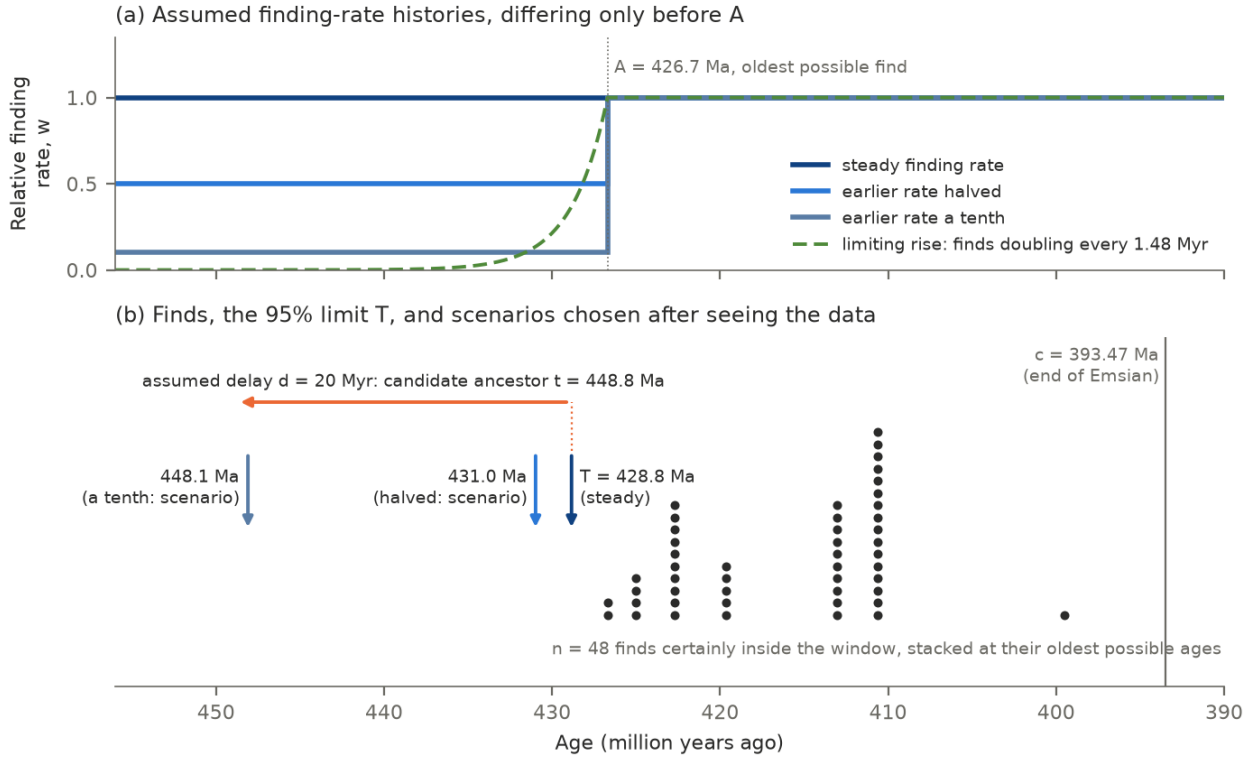


Figure 2: How the body-fossil limit links fossils to ancestors, shown with the main crown vascular-plant finds. (*a*) The finding rate is steady after the oldest find; before it, the rate is unchanged, halved, cut to a tenth, or rises exponentially at the limiting rate for 48 finds, doubling every 1.48 Myr: that rise keeps the summed earlier finding within the equivalent of 2.1 Myr at the later rate, so it fits an ancestor of any age. It is a requirement for fitting. (*b*) Dots show the 48 finds certainly inside the window, at the oldest ends of their age ranges and stacked for visibility. For an assumed finding-rate history, the oldest find A and the count n give the 95% confidence limit T on the start of the recognisable record. For finds dated by age ranges, A is the oldest possible age of the oldest find and n counts only finds certainly inside the window. In the scenarios chosen after seeing the data, halving the earlier finding rate gives a compatibility endpoint of 431.0 Ma, and cutting it to a tenth gives 448.1 Ma; the steady-rate 95% confidence limit is 428.8 Ma. An assumed delay d before recognisable bodies links the start of the record to a candidate ancestor age t; the arrow adds a 20 Myr delay to the steady-rate limit, giving 448.8 Ma.

Fig. 3.

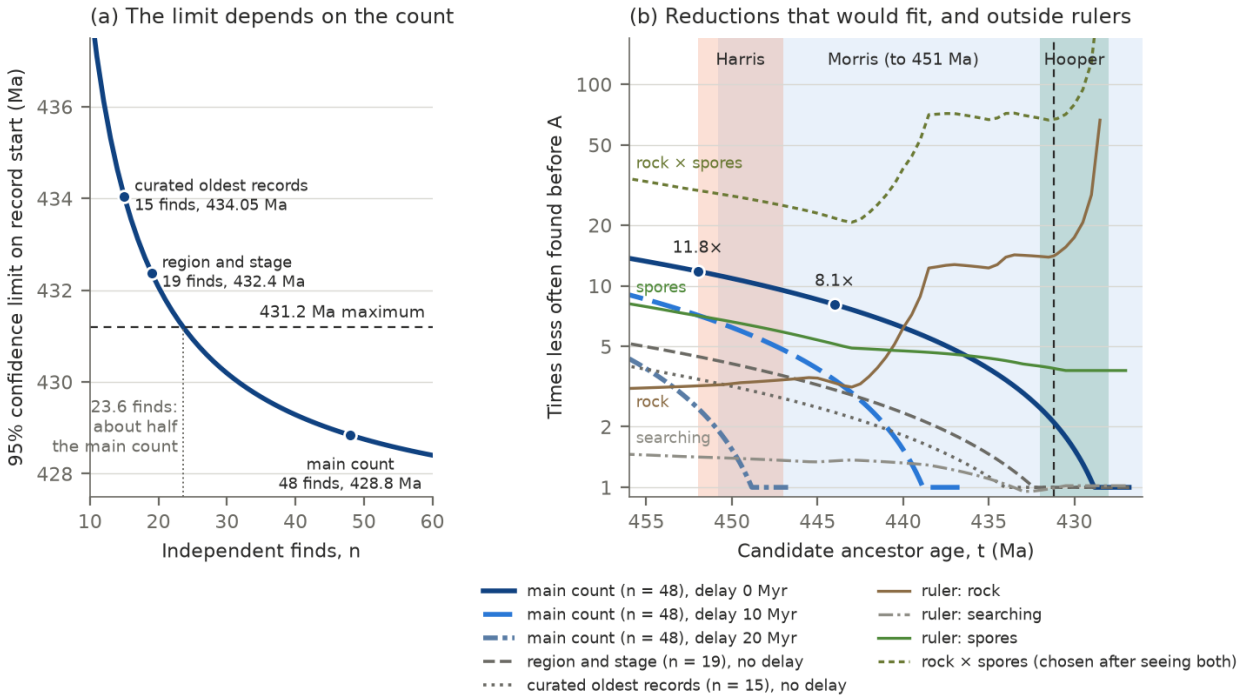


Figure 3: What older vascular-plant ancestors require of the fossil record. (*a*) The 95% confidence limit on the start of the crown vascular-plant record at a steady finding rate, against the number of independent finds n, for the oldest find $A = 426.7$ Ma and the younger cutoff $c = 393.47$ Ma. Dots mark the main count of 48 finds certainly inside the window (428.8 Ma), region–stage cells (19; 432.4 Ma) and the curated oldest records (15; 434.05 Ma). The curve crosses the 431.2 Ma maximum (dashed) at 23.6 finds, about half the main count. (*b*) How many times less often fossils would have to have been found before the oldest find than after it, for each candidate ancestor age t. The finding rate is steady after A, the oldest possible age of the oldest find. Where a curve rises above one, an oldest find no older than A has probability 0.05 given the count, the rule behind the steady-rate limit, with the record starting at $t - d$. Main curves use $n = 48$ and assumed delays d of zero, ten and twenty Myr before recognisable bodies; zero-delay comparison curves count by region and stage ($n = 19$) or only the curated oldest records ($n = 15$). Shading marks the published clock intervals; Morris's younger interval continues outside this panel (figure 1). Thin lines show three outside measures of how many times scarcer something was before A than after it, for an ancestor at t with zero delay: area of North American columns holding non-marine units, PBDB land and marginal-marine collections, and the share of indexed Palynodata studies reporting trilete spores; they are rough guides without confidence status and compare only with the zero-delay curves. The dotted olive line multiplies the rock and spore measures stage by stage, a combination chosen after both were seen. The rock measure is undefined for candidates younger than 428.25 Ma, where the earlier interval holds no qualifying non-marine units in the recorded columns, and runs off the panel for slightly older candidates. Combinations with $t - d \leq A$ are omitted; flat portions at one show ages a steady finding rate already allows. We placed the step after seeing the data, so the curves show what would fit and carry no 95% confidence. Older ancestors fit if early fossils were found several times less often, or recognisable bodies appeared later.